Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering
09/2023 – Expected 06/2028

- GPA: 2.93/4.00
- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing), doing wakeboarding, kiteboarding, surfing, bouldering, running, fitness, and traveling as hobbies.
- Volunteer: Volunteering Division Member at Industrial Engineering Club (BÜYAK); provided educational mentorship to high school students.

Kabataş Boys' High School
09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.
- Volunteer: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX
09/2026 – Present

- Joining the backend development team to contribute to scalable enterprise software solutions, robust backend architecture, RESTful API design, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Intern at Akbank
07/2026 – 09/2026

- Completed a 2-month internship with the SRE team, contributing to the reliability, scalability, and performance of mission-critical banking systems. Engaged in Linux systems engineering, operational monitoring, and high-availability systems management.

Software Engineering Intern at Smash Mate
06/2026 – 07/2026

- Completed an intensive 1-month internship during the critical pre-launch phase, contributing to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native). Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Played an active role in agile workflows, code reviews, and system optimizations, leading directly to the product's successful official launch.

Technical Projects

- **AlphaTrigger** — Autonomous trading simulator with Z-Score mean reversion & volatility sizing.
- **FinTech Platform** — Real-time dashboard with limit-order matching engine & portfolio risk analytics.
- **Seated** — Multi-tenant SaaS for real-time restaurant floor plans, reservations & operations.
- **Performo** — Administrative console with role-based access control (RBAC) & task workflows.
- **Enterprise FAQ** — Enterprise knowledge base with optimistic UI updates, caching & error boundaries.
- **Space Shooter IPC** — Asynchronous 60 FPS input engine using non-blocking FIFOs and signals.
- **SchedSim** — Low-level CPU scheduling simulator executing FCFS, SJF, SRTF, and Round Robin.
- **Solar Escape** — Bare-metal pathfinding algorithm with greedy route selection & manual stack framing.
- **CoreMark Profiling** — Benchmarking study assessing compiler optimizations and ISA pipeline throughput.
- **Spireslayer** — Unix command interpreter featuring a custom backtracking recursive descent parser.
- **Aura** — Functional programming language interpreter supporting closures, lexical scoping, and currying.
- **MatrixNet** — Network topology analysis tool implementing Tarjan's SCC algorithm & articulation points.
- **GigMatch-Pro** — Algorithmic job-matching engine leveraging max-heaps for priority-based candidate allocation.
- **Smash Minesweeper** — Mobile puzzle game built with reducer-driven state engine and BFS flood-fill.
- **Qt 2048** — Cross-platform game compiled to WebAssembly for browser deployment with state undo.
- **Nightpass** — Terminal-based strategy survival card game driven by a deterministic finite state machine.

Skills, Tools & Languages

- **Programming Languages:** Python, Java, C (C11), C++ (C++17), TypeScript, JavaScript, SQL, x86-64 Assembly, RISC-V Assembly (RV32I)
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Tools & Software:** Git/GitHub, Linux/POSIX, AI-Assisted Development, MS Office
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)